

The Impact of High Blood Pressure on Heart Failure Events: A Propensity Score Matching Analysis

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OBJECTIVES

- To evaluate the isolated causal impact of high blood pressure on the occurrence of heart failure events.
- To mitigate selection bias and effectively balance baseline clinical characteristics between patient groups using Propensity Score Matching (PSM).

METHODOLOGY

- A retrospective observational dataset comprising 25,000 patient records obtained from the "Heart Failure Clinical Records" [1] dataset on Kaggle.
- Treatment Variable (T): High blood pressure status
- Outcome Variable (Y): Occurrence of a heart failure event
- Covariates (X): Baseline confounders including age, sex, cholesterol, maximum heart rate, ST depression (oldpeak), fasting blood sugar, exercise-induced angina, chest pain types, resting ECG, and ST slope

Result

Table 1 Characteristics of Heart Failure Patients

Characteristics	Normal BP	High BP	Total	P
N (%)	17,344 (69.40)	7,656 (30.60)	25,000 (100.00)	
Gender, N (%)				0.805
Male	7,782 (44.87)	3,509 (45.83)	11,291 (45.16)	
Female	9,562 (55.13)	4,147 (54.17)	13,709 (54.84)	
Age (years), Median (IQR)	57 (49 – 66)	58 (49 – 65)	58 (49 – 66)	0.655
Cholesterol (mg/dL), Median (IQR)	240 (203 – 277)	239 (202 – 277)	240 (203 – 277)	0.268
Max Heart Rate (bpm), Median (IQR)	162 (146 – 178)	162 (146 – 178)	162 (146 – 178)	0.421
Oldpeak, Median (IQR)	0.80 (0.30 - 1.60)	0.80 (0.30 - 1.50)	0.80 (0.30 - 1.50)	0.216
Heart Failure, N (%)				<.001
Yes	9,845 (56.76)	4,640 (60.61)	14,485 (57.94)	
No	7,499 (43.24)	3,016 (39.39)	10,515 (42.06)	
Fasting blood sugar (>120 mg/dL), n (%)				0.578
Yes	3,788 (21.84)	1,648 (21.53)	5,436 (21.74)	
No	13,556 (78.16)	6,008 (78.47)	19,564 (78.26)	
Exercise-induced chest pain				0.522
Yes	7,000 (40.36)	3,123 (40.79)	10,123 (40.49)	
No	10,344 (59.64)	4,533 (59.21)	14,877 (59.51)	
Chest Pain				0.189
Asymptomatic	8,683 (50.06)	3,767 (49.20)	12,450 (49.80)	
Atypical Angina	3,228 (18.61)	1,403 (18.33)	4,631 (18.52)	
Non-Anginal Pain	3,816 (22.00)	1,712 (22.36)	5,528 (22.11)	
Typical Angina	1,617 (9.32)	774 (10.11)	2,391 (9.56)	
Resting ECG				0.049
LVH	3,458 (19.94)	1,597 (20.86)	5,055 (20.22)	
Normal	10,439 (60.19)	4,482 (58.54)	14,921 (59.68)	
ST	3,447 (19.87)	1,577 (20.60)	5,024 (20.10)	
ST Slope				0.442
Down	1,673 (9.65)	753 (9.84)	2,426 (9.70)	
Flat	7,804 (45.00)	3,497 (45.68)	11,301 (45.20)	
Up	7,867 (45.36)	3,406 (44.49)	11,273 (45.09)	

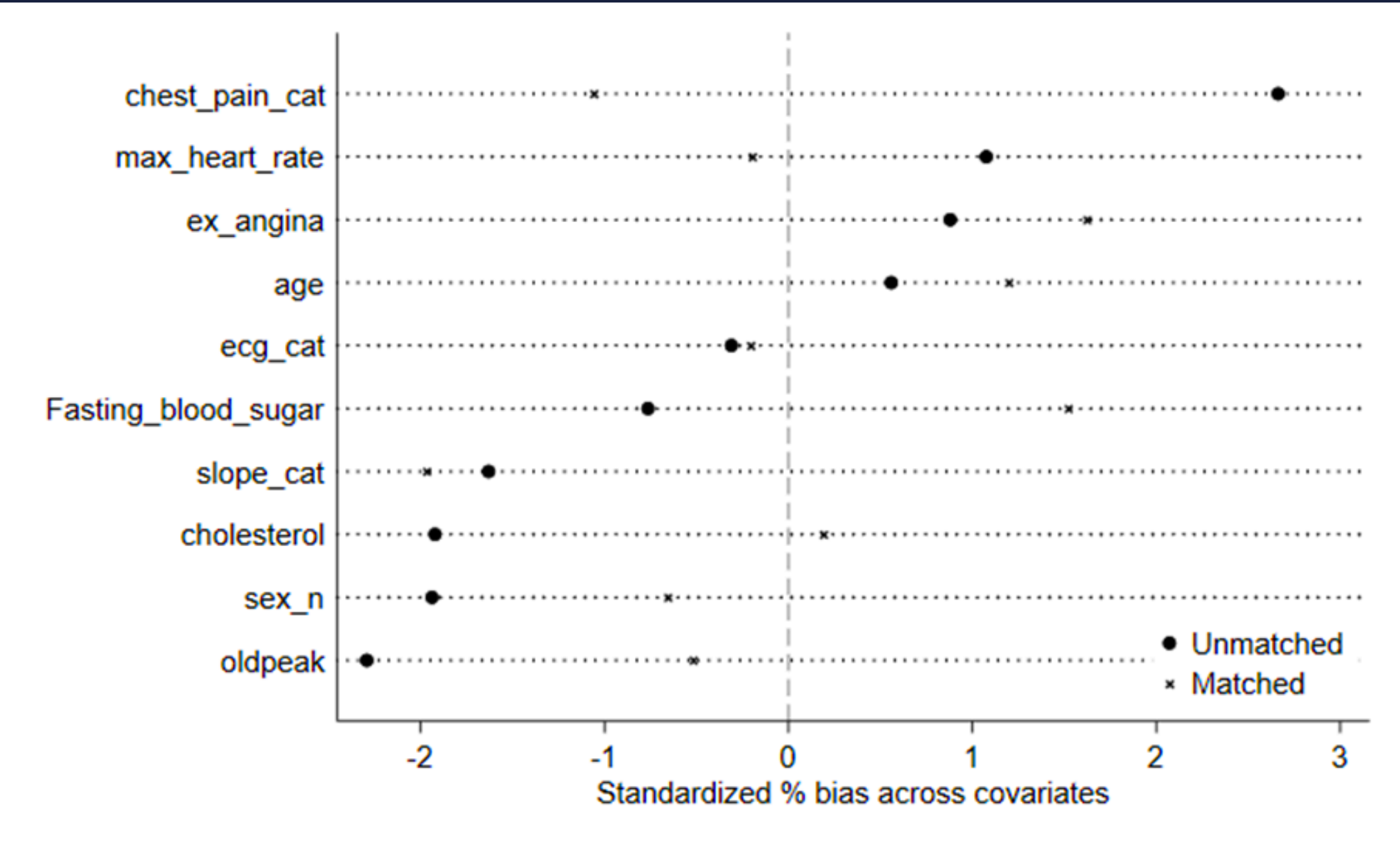


Figure 1 Standardized percentage bias across clinical covariates before (Unmatched) and after (Matched) propensity score matching.

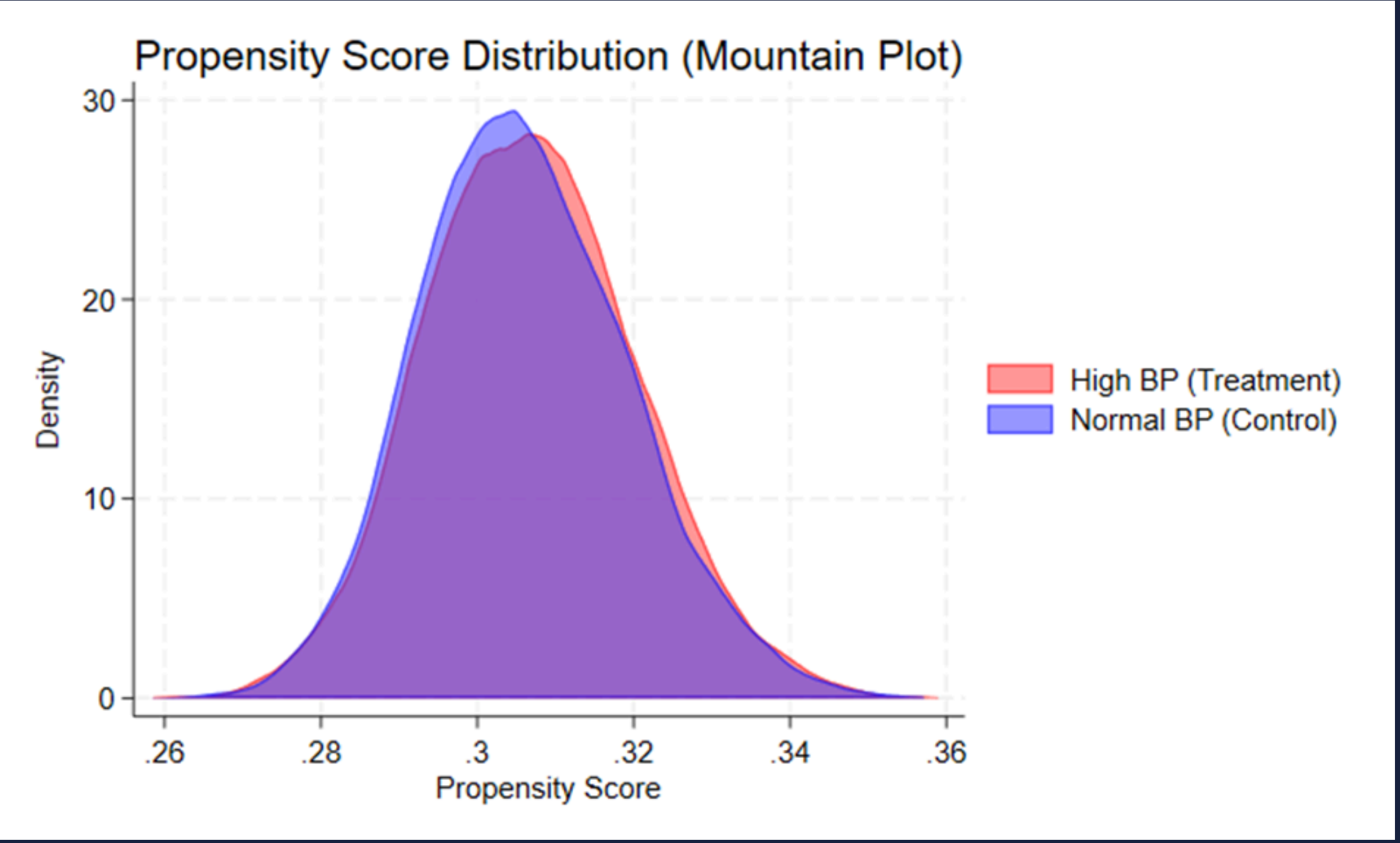


Figure 2. Propensity score distribution and common support region between the high blood pressure (Treatment) and normal blood pressure (Control) groups.

Table 2 Estimated Effect of Hypertension on Heart Failure Event (ATT)

Variable	Sample	Treated	Controls	Difference	S.E.	t
Heart Failure	Unmatched	0.6061	0.5676	0.0384	0.0068	5.68
	ATT	0.6061	0.564	0.0421	0.0092	4.58

CONCLUSIONS

This study confirms that high blood pressure is a significant independent predictor of heart failure. By using propensity score matching to eliminate selection bias, we achieved a fair and reliable comparison between groups. The results demonstrate that patients with high blood pressure have a statistically significant 4.21% higher risk of experiencing heart failure events compared to those with normal blood pressure.

REFERENCE

- [1] Lakhani, H. (n.d.). Heart Failure Clinical Records (25k patients). Kaggle.
- [2] Leuven, E., & Sianesi, B. (2003). PSMATCH2: Stata module to perform full Mahalanobis and propensity score matching.